

19TH DECEMBER 2025/26

Gesture-Controlled Excel Interaction System

Multimodal Interaction

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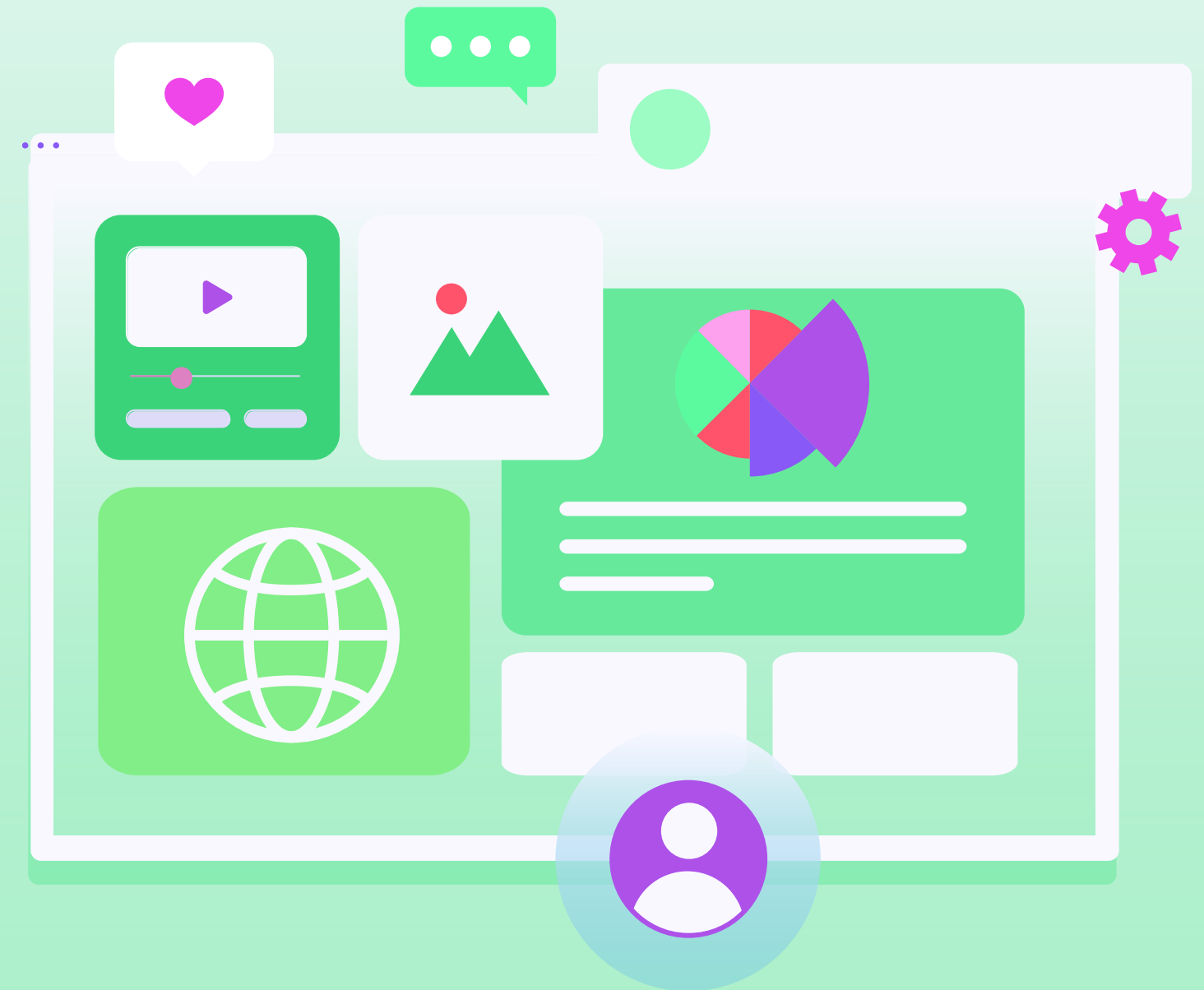




TABLE OF CONTENTS



01 Introduction

- Why we decided to develop this project
- Persona & Scenario
- Goals

02 Architecture

- Interaction of different modules

03 Implemented Intents - Gestures

- How the user can interact with the application

04 Demo

05 Why These Gestures?

06 Number of Annotated Examples

07 Challenges & Conclusion

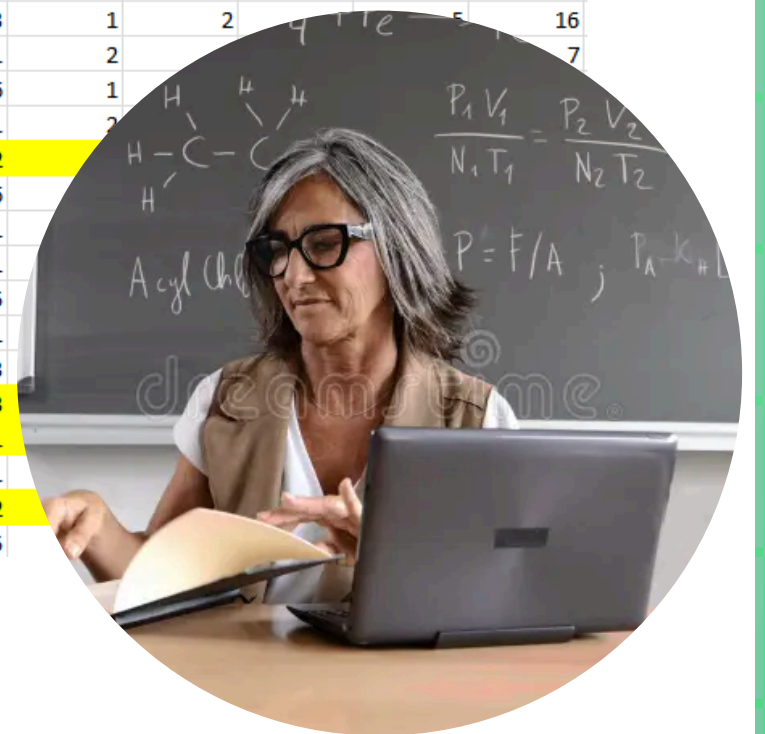
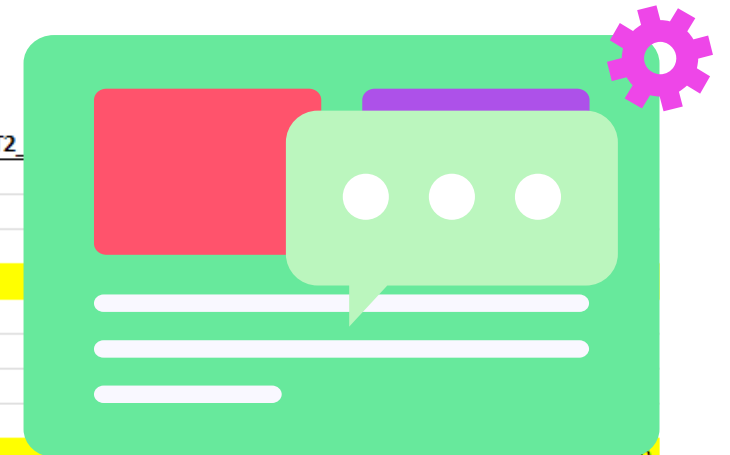
- What challenges did we face

01 Introduction

Persona

- **Name:** Prof. Helena Martins
- **Age:** 58 years old
- **Field:** Social Sciences / Education
- **Experience:** 30 years of university teaching
- **Digital Skills:** Intermediate (uses email, Moodle, and PowerPoint, but avoids advanced tools)

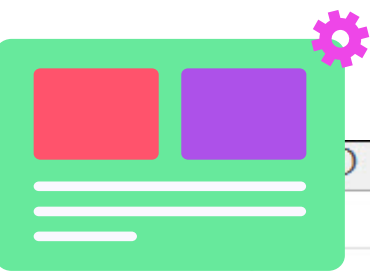
matrícula	Nome	preço de e-	digito de cul	TP	P	REGIME	Teste 1	T2						
107447	MARIA INÉ	maria.ines	8314	TP2	P1a	O	13							
94716	ANA CATA	acatarinafi	8246	TP1	P8	O	8							
107135	MARIANA	marianam	8314	TP2	P2	O	11							
107390	MARIA JO	mariaforte	8314	NO_TP			11							
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107440	PEDRO GA	gab.olivei	8314	TP2	P1	O	16	5	1	2	5	2	15	
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107259	DANIEL FIL	daniel.mo	8314	TP1	P10	O	9	2						
100088	MARIA MI	maria.cast	8307	TP2	P1	O	13	6						
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110376	RICARDO C	rfigueired	8314	TP2	P1	O	16	1						
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108694	MIGUEL X	miguelxlir	8314	TP1	P10	O	11	1						
108421	AFONSO R	marques.a	8314	TP2	P1	O	13	1						
108366	LEONOR M	leonor.ma	8314	TP1	P10	O	8	2						
109462	JOÃO HEN	joaohenric	8314	TP1	P10	O	10	5						



01 Introduction

Persona

- **Difficulty using advanced Excel features**
 - Difficulty calculating averages
 - Difficulty applying conditional formatting
 - Difficulty generating charts
 - Difficulty creating pivot tables
 - Particularly struggles with writing and understanding formulas
- **Fear of causing errors in the spreadsheet**
 - She feels insecure when working in Excel, mainly because she is afraid of accidentally deleting important data or altering the layout or structure of the sheet,



	A	B	C
1		Jan	
2	Entertainment		
3	Cable TV	52.98	52.98
4	Video Rentals	7.98	11.97
5	Movies	16.00	32.00
6	CDs	18.99	29.99
7	Totals	=SUM(B3:B6)	
8			



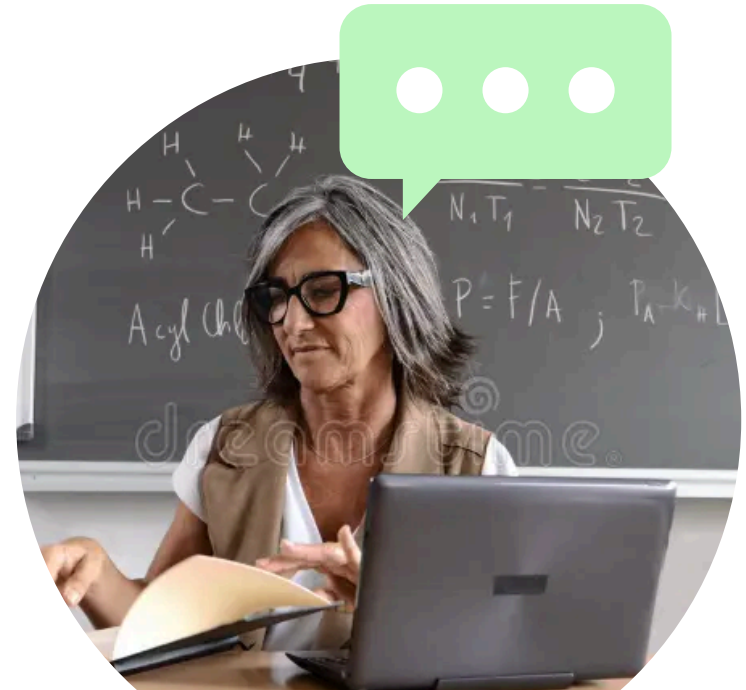
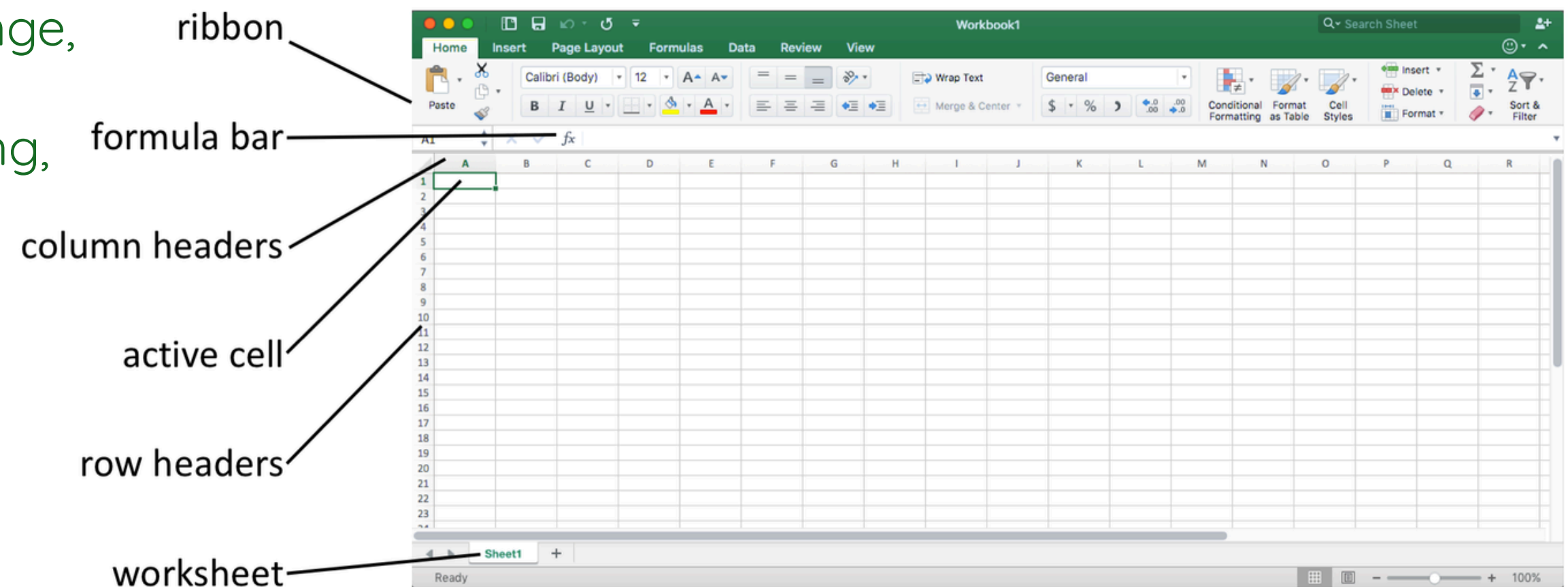
01 Introduction

Persona

- **Difficulty understanding Excel's technical vocabulary**

Common Excel terminology is not intuitive for her, including:

- column, row, cell, range,
- pivot table,
- conditional formatting,
- bar chart.



01 Introduction

Goals



- **1. Automate common Excel tasks using natural gestures**
Use gestures instead of the mouse and keyboard to perform Excel actions.
- **2. Improve task execution efficiency through gestural control**
Complete repetitive Excel tasks faster with gestures.
- **3. Reduce manual navigation via natural interactions**
Control Excel without touching the computer
- **4. Support hands-free spreadsheet interaction**
Control Excel without touching the computer, ideal for teaching and presenting.
- **5. Increase accessibility through gesture-based commands**
Help users with little Excel or digital experience use Excel more easily.
- **6. Reduce user errors through controlled gesture**
Reduce mistakes by using predefined gesture commands.



01 Introduction

Scenario

Context

- Professor Helena Martins is preparing students' final grades in Excel, calculating averages and reviewing results before submission.

Interaction

- Because the office must remain quiet and she feels insecure using advanced Excel features, she uses a gesture-based Excel control system with a Kinect sensor. Through simple, predefined gestures, she can calculate averages, apply formatting, and move between sheets without using the mouse or keyboard.

Outcome

- The gesture-based interaction allows her to work silently, reduce errors, and complete the task more efficiently and confidently.



02 System Architecture Overview

User



Processing Layer

Gesture Recognition Engine
(Visual Gesture Builder)



Fusion Engine
(MMI Framework)

Combines gesture events
Maps gestures to application
commands

Excel

Microsoft Excel

Controlled via Excel Interop
Executes spreadsheet actions
Provides visual feedback to the user

Input Layer

Sensor – Kinect XBOX



Application Layer

Backend (C#)



User → Kinect → Gesture Recognition → Fusion Engine → Backend → Excel



03 Implemented Intents - Gestures

Excel Voice Commands vs. Gesture Commands



Voice Commands

- Calculate average
- Highlight passed and failed students
- Insert columns
- Apply real improvement
- Apply possible improvement
- Perform mathematical operations
- Insert questions
- Generate class performance charts
- Generate student bar charts
- Generate question-based charts
- Delete all charts
- Save file / report
- Update grades
- Create pivot table
- Display help

Gestures Commands

- Zoom in
- Zoom out
- Navigate up
- Navigate down
- Navigate left
- Navigate right
- Calculate average
- Insert column
- Highlight passed students
- Highlight failed students
- Clear cell content
- Close Excel

12

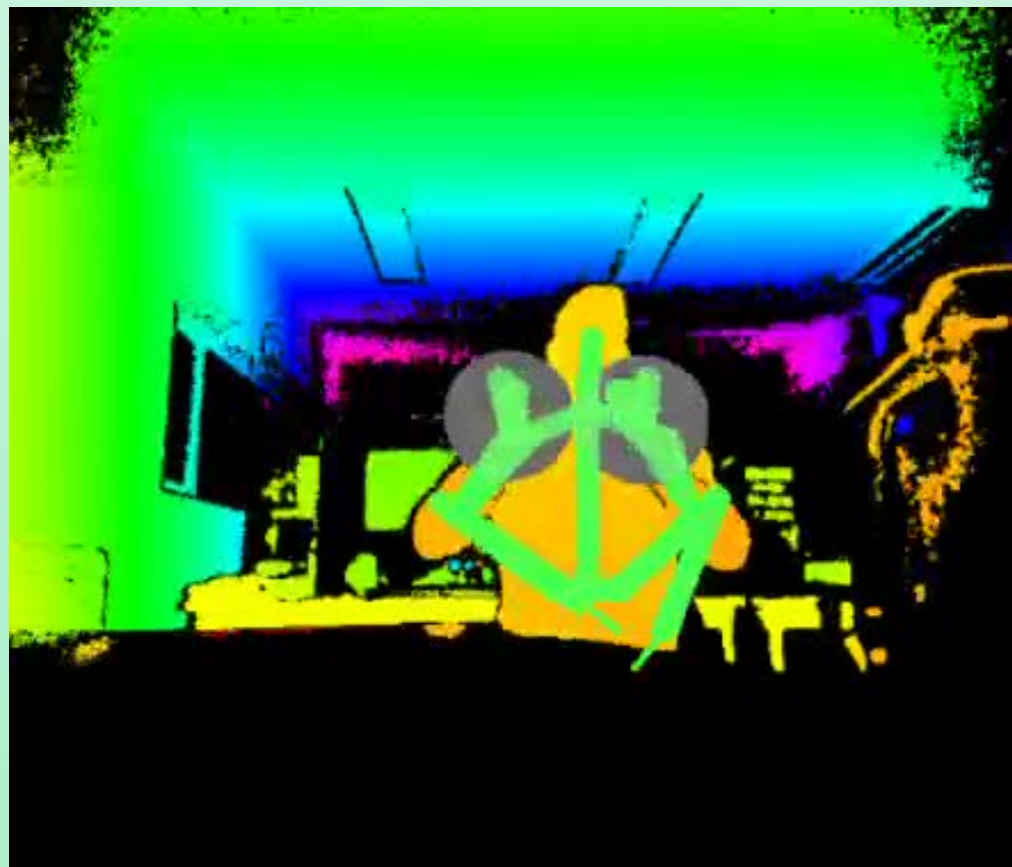




03 Implemented Intents - Gestures

Explanation: Facilitates data readability without manual navigation

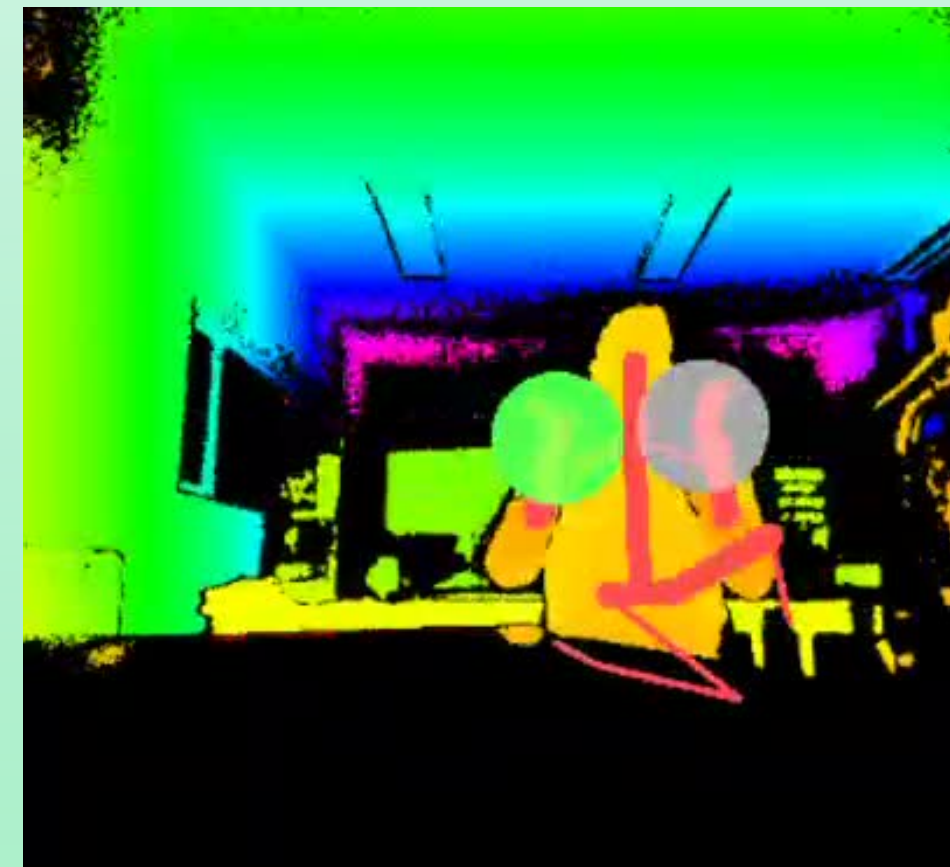
Zoom In



Excel Action

Increase spreadsheet zoom

Zoom out



Excel Action

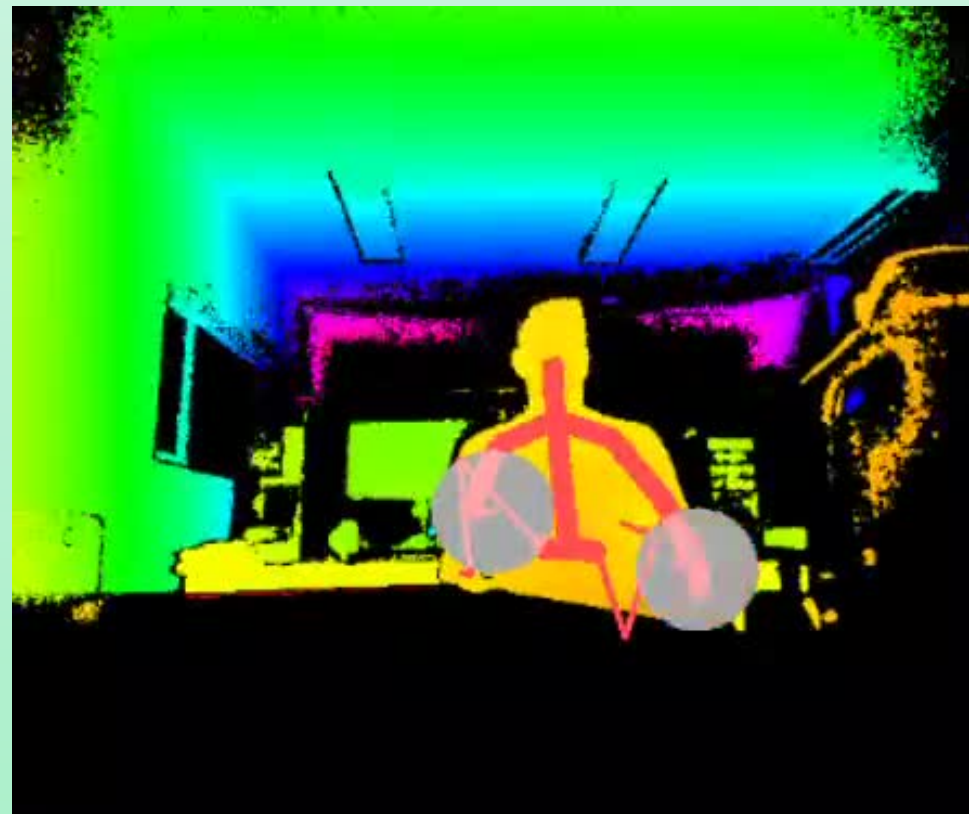
Decrease spreadsheet zoom



03 Implemented Intents - Gestures

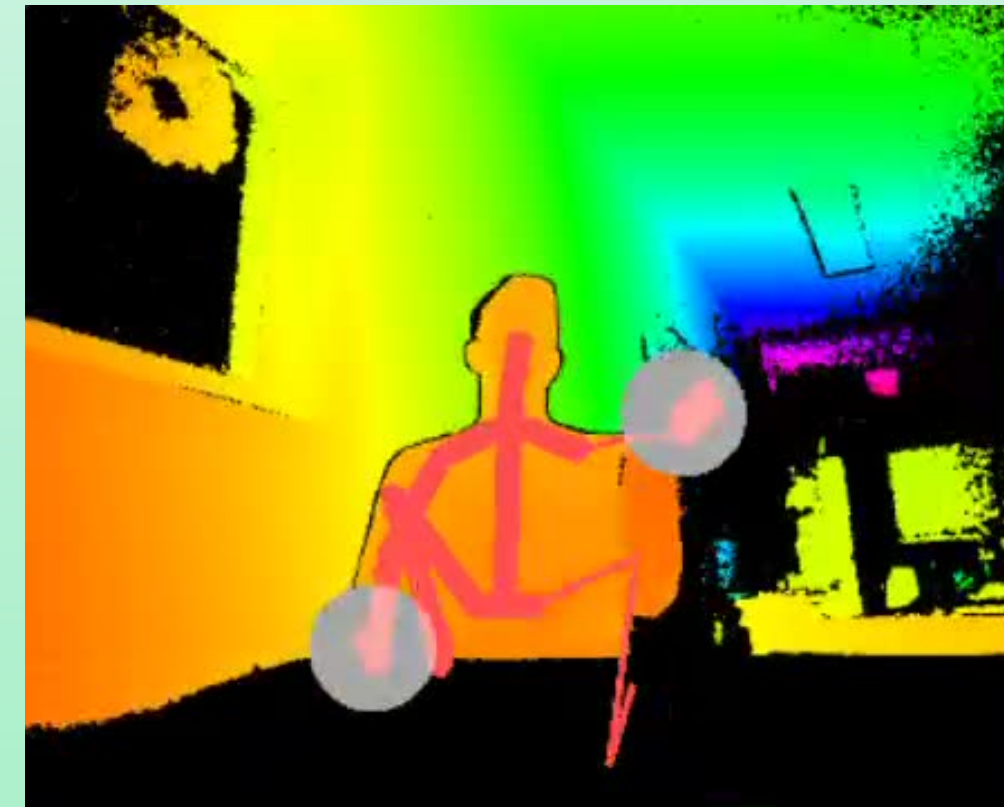
Explanation: Allows quick navigation across spreadsheet areas

Swipe Down



Excel Action
Navigate down

Swipe Up



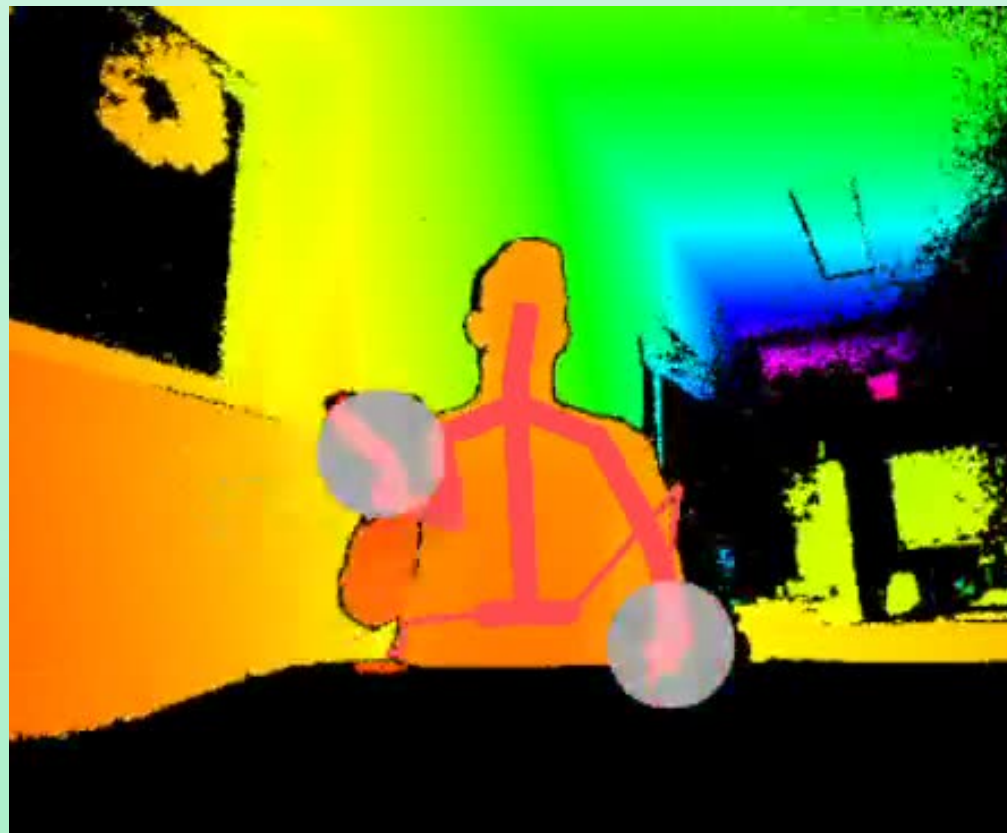
Excel Action
Navigate up



03 Implemented Intents - Gestures

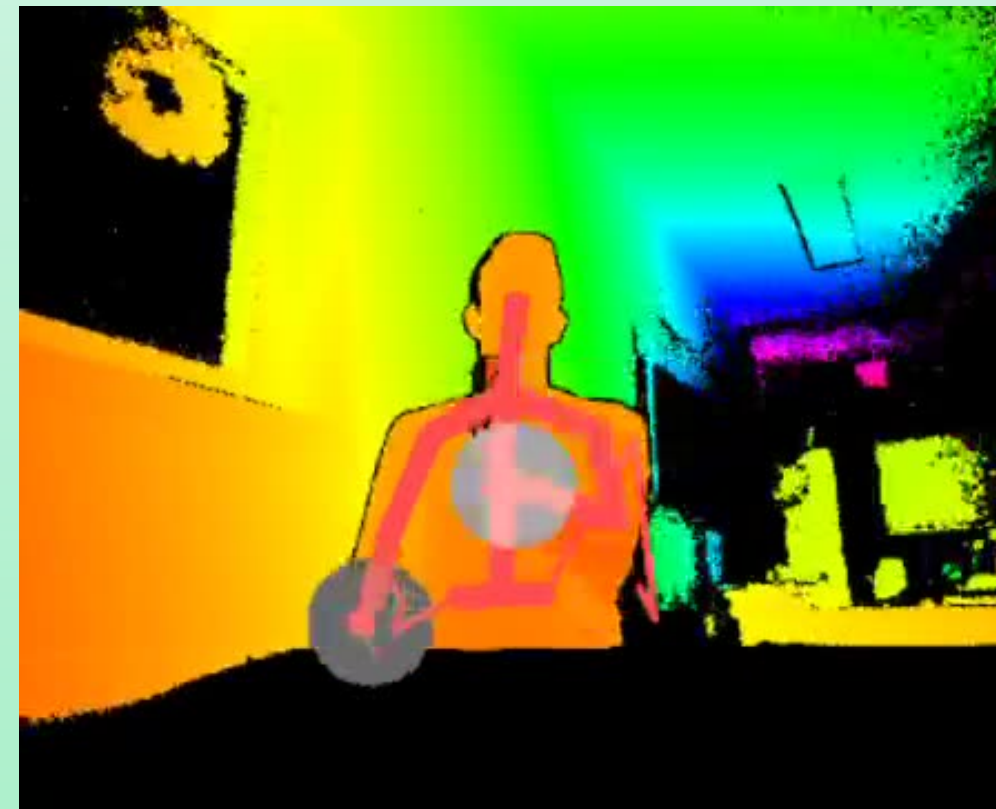
Explanation: Allows quick navigation across spreadsheet areas

Swipe Right



Excel Action
Navigate left

Swipe Left



Excel Action
Navigate right



03 Implemented Intents - Gestures

Explanation: Supports visual analysis of results

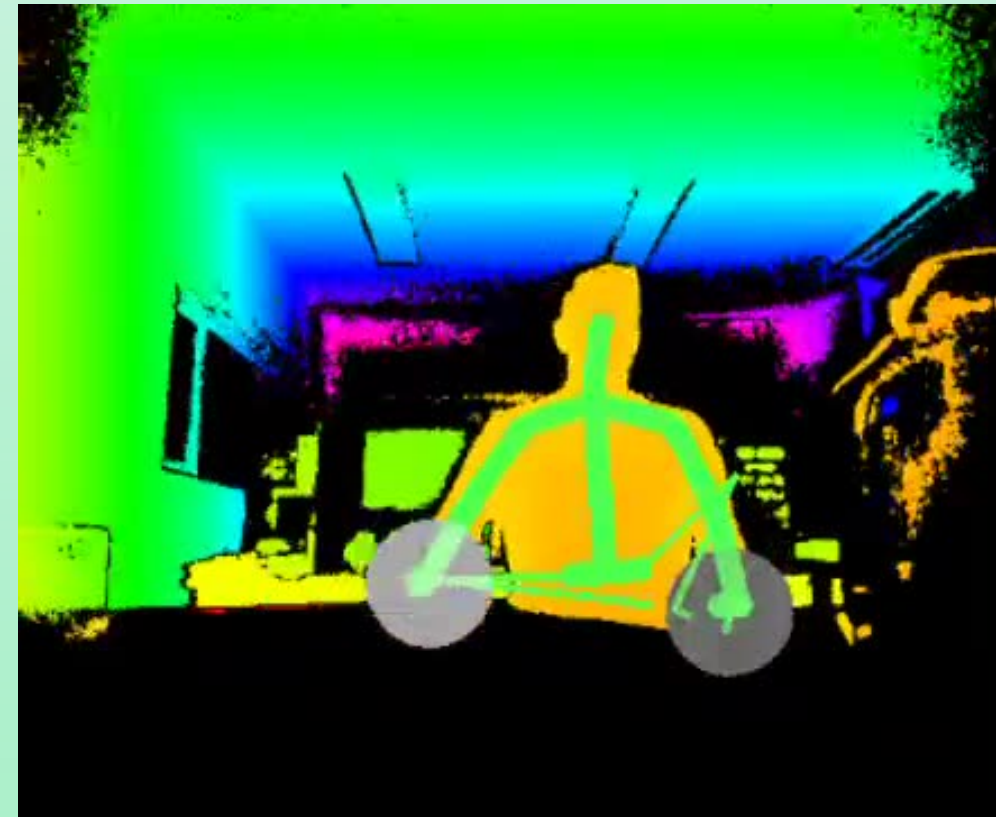
Highlight Failed



Excel Action

Highlight falling values

Highlight Passed



Excel Action

Highlight passing values

03 Implemented Intents - Gestures

Explanation: Allows error correction without direct cell manipulation

Delete Cell Value



Excel Action

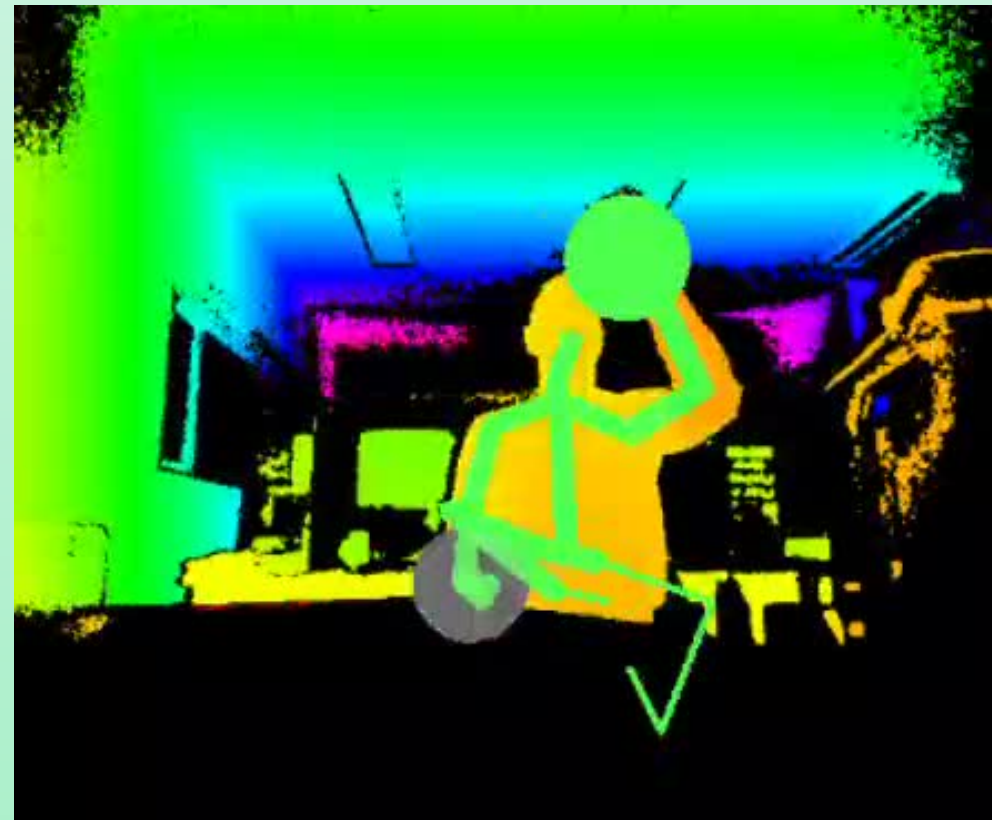
Clear selected cell content



03 Implemented Intents - Gestures

Explanation: Automates a common calculation without requiring formulas

Calculate Average



Excel Action

Calculate average of the selected cells

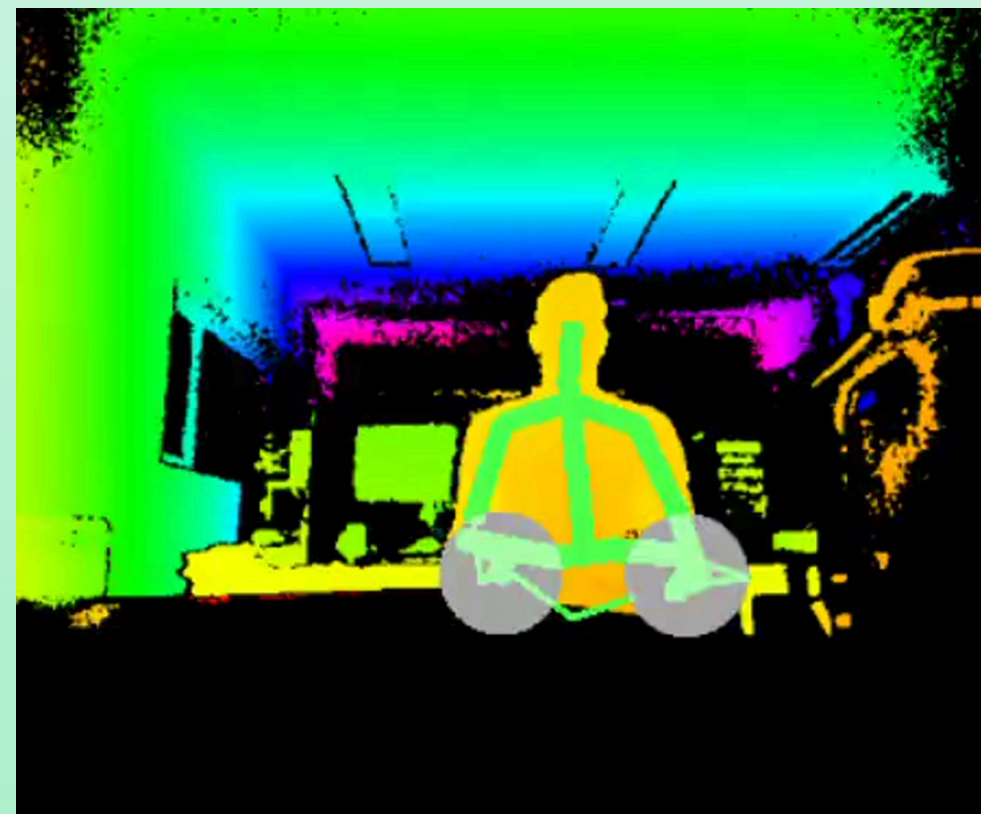


03 Implemented Intents - Gestures



Explanation: Enables simple and efficient data organization

Insert Column



Excel Action

Insert a new column



03 Implemented Intents - Gestures



Explanation: Closes the application quickly and safely

Close Excel



Excel Action

Close the Excel application



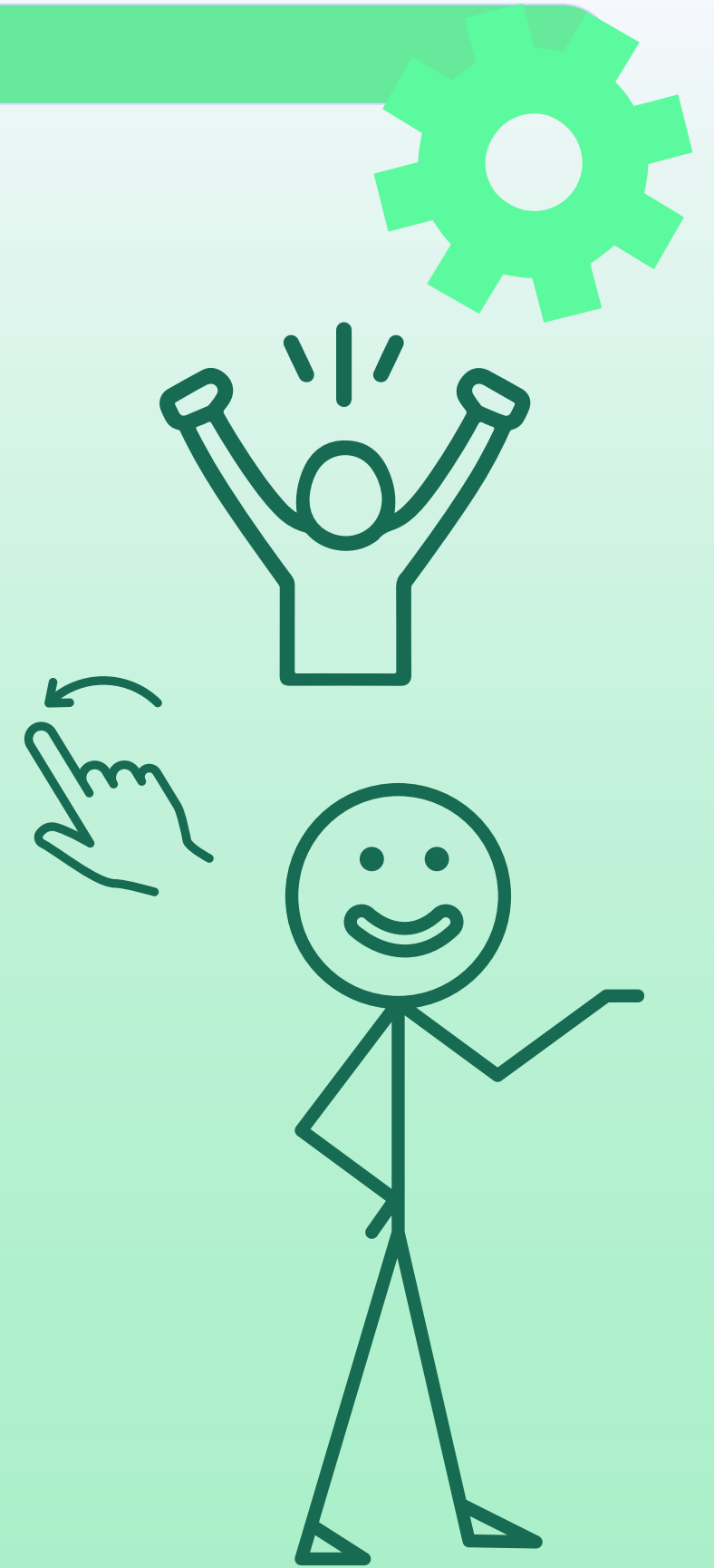
04 DEMO





05 Why These Gestures?

- The selected gestures are **simple, natural, and easy to learn**.
- Each gesture has a **clear and intuitive mapping** to its corresponding Excel action.
- Simple and distinct gestures reduce recognition errors and false positives.
- Gesture-based interaction **supports silent, hands-free use in shared environments**.
- The gestures **focus on frequent and essential Excel tasks**, increasing efficiency and user confidence.





06 Number of Annotated Examples

Training Process for Each Gesture





06 Number of Annotated Examples

3 Calculate Average 17

3 Zoom In 10

3 Zoom Out 16

2 Close Excel 9

3 Insert Column 9

3 Swipe Right 18

4 Highlight Passed 11

4 Highlight Failed 13

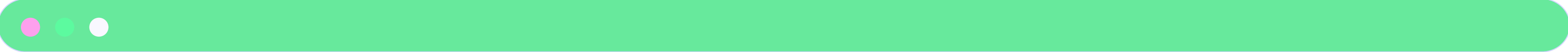
2 Swipe Up 9

2 Swipe Down 9

3 Swipe Left 11

2 Delete Cell Value 7

CONFIDENCE: 0.6



07 Challenges & Conclusion

Challenges

- **1. Gesture recognition accuracy**
Avoiding false positives and missed gestures..
- **2. Gesture differentiation**
Designing gestures that are intuitive and non-conflicting
- **3. User variability**
Handling different body sizes, speeds, and movement styles.
- **4. Real-time responsiveness**
Ensuring low latency between gesture and Excel action.
- **5. Complex Interconnection of All Components**
Coordinating Kinect, gesture recognition, Fusion, and Excel backend.



07 Challenges & Conclusion

Conclusion

- Gesture interaction makes **Excel** more **intuitive** and **accessible**.
- Reduces dependence on menus and complex commands.
- Enables **hands-free interaction** for common Excel tasks.
- System successfully executes Excel actions using natural gestures.
- **Future work:** add more gestures and combine gestures with voice.



THANK YOU!

Voice-Controlled Excel Management System

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